

Seat
No.

PD-03

Total No. of Pages : 12

Shivaji University, Kolhapur
Ph.D. Entrance Examination, February- 2024

COMPUTER SCIENCE

Sub. Code : 58802

Day and Date : Wednesday, 07-02-2024

Total Marks : 100

Time : 01:00 pm to 03:00 pm

Instructions:

- 1) All questions are compulsory.
 - 2) Each question carries 2 marks.
 - 3) Answers should be marked in the given OMR answer sheet by darkening the appropriate option.
 - 4) Use black ball point pen only for marking the circle. Do not make any stray mark on the OMR Answer Sheet.
 - 5) Follow the instructions given on OMR Sheet.
 - 6) Rough work shall be done on the sheet provided at the end of question paper.
 - 7) Only non programmable calculators are allowed.
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SECTION-1

Q1. Basic research is also known as_____

- a) Pure research
- b) Fundamental research
- c) Theoretical research
- d) All the above

Q2. What is a possible explanation that we may not agree with?

- | | |
|----------------------|-----------------------|
| a) Research question | b) Research Objective |
| c) The Hypothesis | d) Grounded Theory |

- Q3. A statement whose validity is tested on the basis of a sample is called?
- a) Null Hypothesis
 - b) Statistical Hypothesis
 - c) Simple Hypothesis
 - d) Composite Hypothesis
- Q4. A research paper _____
- a) Is a compilation of information on a topic
 - b) Contains original research as deemed by author
 - c) Contains peer-reviewed original research or evaluation of research conducted by others
 - d) Can published in more than one journal
- Q5. A statement made about a population for testing purpose is called?
- a) Statistic
 - b) Hypothesis
 - c) Level of Significance
 - d) Test-Statistic
- Q6. _____ is a quality of a good researcher
- a) Age
 - b) Time
 - c) Scientific temper
 - d) Money
- Q7. The issue of Research ethics may be considered pertinent at which stage of research
- a) at the stage of problem formulation and its definition
 - b) at the stage of defining the population of research
 - c) at the stage of data collection and interpretation
 - d) at the stage of reporting the findings.

- Q8. The basis components in legal reasoning is are _____
- a) logic
 - b) justice
 - c) experience
 - d) all of these
- Q9. _____ is defined as the square-root of the average of squares of deviations
- a) standard deviation
 - b) mean deviation
 - c) variance
 - d) none of these
- Q10. _____ is the determination of a statistical relationship between two or more variables.
- a) Regression
 - b) Correlation
 - c) Association
 - d) none of these
- Q11. We review the relevant literature to know _____
- a) What is already known about the topic
 - b) What concepts and theories have been applied to the topic
 - c) Who are the key contributors to the topic
 - d) All of the above
- Q12. Amongst which of the following step is performed by data scientist after acquiring the data?
- a) Deletion
 - b) Data Replication
 - c) Data Integration
 - d) Data Cleansing
- Q13. Problems that can be solved in polynomial time are known as
- a) intractable
 - b) tractable
 - c) decision
 - d) complete

- Q14. Stochastic optimization refers to _____
- a) the use of randomness in the objective function or in the optimization algorithm.
 - b) where the data for the given problem is known accurately
 - c) the use of linear function in the objective function or in the optimization algorithm
 - d) None of these
- Q.15. An algorithm which uses the past results and uses them to find the new result is _____
- a) Brute Force
 - b) Divide and conquer
 - c) Dynamic programming algorithm
 - d) None of these
- Q.16 Data often contain
- a) Target Class
 - b) Uncertainty
 - c) Methods
 - d) Keywords
- Q.17 You want to minimize development cost by reusing methods? Which design pattern would you choose?
- a) Adapter Pattern
 - b) Singleton Pattern
 - c) Delegation pattern
 - d) Immutable Pattern

- Q.18 _____ is held in the catalog of the warehouse database system
- a) Application level metadata
 - b) Algorithmic level metadata.
 - c) Departmental level metadata.
 - d) Core warehouse metadata.
- Q.19 Which of the following algorithm implementations is similar to that of an insertion sort?
- a) Binary heap
 - b) Quick sort
 - c) Merge sort
 - d) Radix sort
- Q.20 Which Cloud Computing service model is most suitable for researchers looking to access virtualized computing resources without managing the underlying infrastructure?
- a) Infrastructure as a Service (IaaS)
 - b) Platform as a Service (PaaS)
 - c) Software as a Service (SaaS)
 - d) Function as a Service (FaaS)
- Q.21 Which technology is commonly used for distributed storage and processing of Big Data?
- a) SQL
 - b) NOSQL
 - c) XML
 - d) Hadoop

- Q.22 In machine learning, what does the term "overfitting" refer to?
- a) Underutilizing the available data
 - b) Fitting the model too closely to the training data
 - c) Generalizing well to new, unseen data
 - d) Achieving optimal model performance
- Q.23 What is the primary purpose of R in the context of statistical computing and graphics?
- a) Image Editing
 - b) Web Development
 - c) Data Analysis and Visualization
 - d) Database Management
- Q.24 Which technology is commonly associated with real-time stream processing in Big Data?
- a) Hadoop
 - b) Spark
 - c) Cassandra
 - d) MongoDB
- Q.25 Which of the following is the application of cloud computing?
- a) Adobe
 - b) Paypal
 - c) Google G Suite
 - d) All of these

SECTION-II

- Q.26. A service that concentrates on hardware follows the _____ as a Service model.
- a) IaaS
 - b) SaaS
 - c) PaaS
 - d) All of the mentioned
- Q.27 Variable name in C language must not start with _____
- a) Digit
 - b) white space character
 - c) Special symbol
 - d) All of these

- Q.28 Defining two methods with the same name but with different parameters is called.
- a) Loading
 - b) Overloading
 - c) Multiplexing
 - d) Overrrding
- Q.29 What is the advantages of MVC?
- a) MVC segregates your project into a different segment, and it becomes easy for developers to work on
 - b) It is easy to edit or change some part of your project that makes project less development and maintenance cost
 - c) MVC makes your project more systematic
 - d) All of the mentioned
- Q.30 The main components of software project life cycle are _____
- a) reviews, software testing, software documentation.
 - b) reviews, requirement gathering, software testing, software documentation
 - c) reviews, expert opinions, software testing, software maintenance
 - d) system requirement specification
- Q.31 If the boundary is specified in a single color, and if the algorithm proceeds pixel by pixel until the boundary color is encountered is known as
- a) Parallel curve algorithm
 - b) Flood-fill algorithm
 - c) Scan-line till algorithm
 - d) Boundary-fill algorithm

- Q.32 Dimensionality reduction reduces the data set size by removing _____
- a) relevant attributes
 - b) irrelevant attributes
 - c) derived attributes
 - d) composite attributes
- Q.33 The _____ is responsible for running queries and reports against data warehouse tables.
- a) hardware
 - b) software
 - c) end users
 - d) middleware
- Q.34 Fractional knapsack problem is solved most efficiently by which of the following algorithm?
- a) Dynamic Programming
 - b) Greedy Algorithm
 - c) Divide And Conquer
 - d) Backtracking
- Q.35 To keep the UI running smoothly, use _____ for long-running tasks, such as all database operations in Android.
- a) coroutines
 - b) ViewModels
 - c) returns
 - d) managed threads

Q.36 What is machine learning?

- a) The autonomous acquisition of knowledge through the use of computer programs.
- b) The autonomous acquisition of knowledge through the use of manual programs.
- c) The selective acquisition of knowledge through the use of computer programs.
- d) The selective acquisition of knowledge through the use of manual programs.

Q.37 An attempt to harm, damage or cause threat to a system or network is broadly termed as _____

- a) Cyber crime
- b) System hijacking
- c) Cyber attack
- d) None of these

Q.38 Identify the major drawback of the Spiral Model?

- a) Does not work efficiently with small projects.
- b) High amount of risk analysis
- c) Strong approval and documentation control
- d) Additional functionalities are added later on

Q.39 _____ is a high level API built on TensorFlow.

- a) PyBrain
- b) Keras
- c) PyTorch
- d) Theano

- Q.40 In Java, what is the role of the "interince" keyword
- a) It declares a class as abstract.
 - b) It defines a blueprint for a class, specifying a set of methods to be implemented.
 - c) It creates an instance of a class.
 - d) It is used to declare variables with constant values.
- Q.41 How does JDBC differ from Hibernate in terms of data access?
- a) JDBC is focused on relational database access, while Hibernate provides a higher- level, object-relational mapping.
 - b) Hibernate is designed for low-level database access, while JDBC provides a more abstract, object-oriented approach.
 - c) JDBC is specific to non-relational databases, while Hibernate supports only relational databases.
 - d) JDBC and Hibernate serve the same purpose and can be used interchangeably.
- Q.42 In Android development, what is the role of the AsyncTask class?
- a) Managing background tasks and long-running operations on a separate titread.
 - b) Defining the structure of the AndroidManifest.xml file
 - c) Handling network communication in Android applications.
 - d) Implementing user authentication and authorization:
- Q.43 Which of the following is a PHP framework commonly used for building web applications and follows the Model-View-Controller (MVC) architectural pattern?
- a) Laravel
 - b) Drupal
 - c) Symfony
 - d) CodeIgniter

- Q.44 In the context of machine learning, what is the purpose of cross-validation?
- a) Testing a model on a completely new dataset.
 - b) Training a model on a large dataset
 - c) Evaluating a model's performance using multiple subsets of data
 - d) Regularizing a model.
- Q.45 Which of the following is true about decorators in Python?
- a) Decorators are used only for defining static methods in a class.
 - b) Decorators are a way to modify or extend the behavior of functions or methods without changing their code.
 - c) Decorators can only be applied to built-in functions, not user-defined functions.
 - d) Decorators are a syntax error and should be avoided in Python.
- Q.46 What is the reusable resource?
- a) that can be used by one process at a time and is not depleted by that use
 - b) that can be used by more than one process at a time
 - c) that can be shared between various threads
 - d) none of these
- Q.47 What is the purpose of the virtual keyword in C++?
- a) It is used to declare a variable with a constant value
 - b) It is used to define a base class.
 - c) It is used to create a constructor in a class.
 - d) It is used to indicate that a function can be overridden in derived classes.

Q.48 SSID stands for _____

- a) Secure Service Identifier
- b) Secure Set Independent Device
- c) Secure Set Identifier
- d) Service Set Independent Device

Q.49 Which normal form is considered adequate for normal relational database design?

- | | |
|--------|--------|
| a) 2NF | b) 5NF |
| c) 4NF | d) 3NF |

Q.50 What is the primary purpose of a firewall in network security?

- a) To prevent physical access to network devices.
- b) To monitor and control network traffic based on predetermined security rules.
- c) To encrypt data transmitted over the network.
- d) To diagnose and troubleshoot network issues

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